

Dry Matter Intake

INTRODUCTION

The control of dry matter intake (DMI) is complex, with mechanisms to ensure an adequate supply of energy as well as to prevent its overconsumption. Various stimulatory and inhibitory signals are integrated in feeding centers of the brain to affect feeding behavior, which determines DMI. Stimulatory signals are related to energy status as well as various sensory, social, circadian, and habitual factors; inhibitory signals include those related to ruminal distension, rumen osmolarity, endocrine effects, and fuel sensing by tissues. Whereas DMI is related to energy required for milk production, maintenance, and change in body reserves, it is also affected by the interaction of diet and physiological state. Understanding factors affecting DMI is essential to optimal ration formulation.

Physiological state and nutrient requirements vary greatly as dairy cattle cycle through lactation. Their diets include forages for proper rumen function, and forage fiber digests and passes from the rumen more slowly than other dietary components. Therefore, DMI by dairy cattle can be limited by distention of undigested residues in the gastrointestinal tract. In the immediate postpartum period, cows are in negative energy balance, but neither the filling effect nor the energy content of rations can be altered to eliminate this. Distention likely begins to control DMI when milk yield (MY) and energy requirements increase in the weeks following parturition and likely limits DMI through peak lactation. As MY declines through lactation, distention usually becomes less limiting, and mechanisms related to specific fuels dominate control of DMI.

PHYSICAL LIMITATIONS

Early investigations of the control of DMI in dairy cattle focused on physical limitations as Lehman (1941) suggested that DMI is limited by the ballast of undigested feed residues

in the gastrointestinal tract. Research related to physical limitations to DMI has focused on forage characteristics that affect gut fill, the site of distention, and the mechanism by which distention limits intake. Feed intake was positively related to forage digestibility with a decreased response as digestibility increased (Blaxter et al., 1961), suggesting that distention from gut fill limits DMI less as digestibility increases. The reticulorumen is generally regarded as the site within the gastrointestinal tract at which distention limits DMI (Campling, 1970; Baile and Forbes, 1974). Tension receptors that respond to distention are located primarily in the reticulum and cranial sac (Leek, 1986). The mechanism by which distention limits DMI is likely via transmission of signals from these tension receptors to brain feeding centers rather than a limitation by volume per se. This is supported by dose-dependent decreases in DMI from additions of inert fill (water- or air-filled bladders, plastic cubes, etc.) into the reticulorumen in several experiments, as reviewed by Allen (1996). In addition, distention in the reticulorumen is determined by both volume and weight of contents; DMI was reduced 112 g for each kilogram of weight and 157 g for each liter of volume that was added to the reticulorumen as inert fill (Schettini et al., 1999). Furthermore, Dado and Allen (1995) reported a reserve volume of 16 L in the reticulorumen of lactating cows consuming a fill-limiting diet, indicating that additional capacity for volume may exist even when distention in the reticulorumen limits DMI.

The abomasum might also be involved in the physical limitation of DMI. Abomasal infusion of methylcellulose decreased DMI by sheep (Grovmum and Phillips, 1978), and distention of the abomasum of sheep with water-filled bladders resulted in a dose-dependent reduction in DMI (Grovmum, 1979). In addition, increasing abomasal emptying rate by suppressing the pyloric gate by pylorotomy and pyloroplasty increased DMI by sheep (Malbert and Ruckebush, 1989). Abomasal distention might generate a satiety signal because mechanoreceptors in the abomasal epithelium have

been described that discharge with distention (Harding and Leek, 1972). Alternatively, restricted flow from the abomasum might limit DMI by decreasing flow from the rumen and increasing ruminal distention. Regardless of the site at which the signal is generated, it is clear that DMI can be limited by distention within the gastrointestinal tract.

The filling effect of forages was reported to be related to fiber mass (Balch and Campling, 1962), and intake of forages was more highly related to their neutral detergent fiber (NDF) concentration than to other chemical measures (Van Soest, 1965). Although NDF concentration has been used as the only ration characteristic to predict DMI by dairy cattle consuming mixed rations of forages and concentrates (Mertens, 1987, 1994), there is substantial evidence that NDF alone is inadequate. Forage NDF has a longer ruminal retention time than other dietary components, including NDF from other sources. Retention time is longer because of longer initial particle size and greater buoyancy in the rumen over time (Allen, 2000). Although most studies report a significant decrease in DMI as forage NDF increased, the DMI response was variable, likely depending on the degree to which intake was limited by undigested feed residues in the rumen and the filling effect of the forage NDF (Allen, 2000).

Increasing ration NDF content by substituting nonforage fiber sources (NFFS) for concentrate feeds has shown little effect on DMI (Allen, 2000). NFFS include by-product feeds such as soyhulls, beet pulp, and distillers grains. Fiber in NFFS is less filling than forage NDF because it is less filling both initially (smaller particle size) and over time because it passes from the rumen more quickly and is often more quickly digested. Therefore, the forage NDF content of rations containing both forages and concentrates is a more important measure related to DMI than total NDF content.

Digestibility of NDF varies among forage type (e.g., grasses versus legumes) and decreases as forages mature and become more lignified. Greater NDF digestibility of forages measured in vitro or in situ was related positively to DMI and MY of lactating cows; a one-unit increase in NDF digestibility corresponded to a 0.17-kg/d increase in DMI and 0.25-kg/d increase in fat-corrected milk (Oba and Allen, 1999b). Within a forage type, NDF that is more fermentable clears the rumen faster and is less filling, allowing greater DMI when limited by distention. However, this applies only within forage type; NDF from perennial cool-season grasses is generally much more digestible than NDF from legumes but is also more filling and more likely to limit DMI, as discussed below.

The filling effect of forage NDF is also affected by initial particle size; decreasing particle size of forages by grinding and pelleting generally increases DMI with a greater intake response by pelleting low-quality forages compared with high-quality forages (Minson, 1963). Long-forage particles are required to form the rumen mat that functions to selectively retain small fibrous particles in the reticuloru-

men, increasing their digestibility (Grant, 1997). However, experiments that have evaluated effects of forage particle size have generally shown only small effects on DMI of lactating cows (Allen, 2000), probably because initial particle size was sufficient to form a rumen mat.

The filling effect of forage NDF is affected by the fragility of forage particles, which affects the rate of reduction in particle size from chewing during eating and ruminating (Poppi et al., 1981). Faster particle size reduction will increase the mass of particles below the threshold size to pass from the reticulorumen as well as decrease the ability of the rumen to selectively retain those particles by decreasing the size of the rumen mat. Although NDF of perennial grasses is less lignified and more digestible than NDF of legumes, DMI is generally lower for perennial grasses than legumes (Oba and Allen, 1999b). Grasses are more filling because retention time in the rumen is greater (Voelker-Linton and Allen, 2008; Kammes and Allen, 2012a,c). Forage fragility is likely a more important factor affecting DMI than in vitro NDF digestibility when comparing legumes and grasses. However, within forage type, in vitro NDF digestibility is likely related to forage fragility. Increased alfalfa maturity decreased passage rate of indigestible NDF and increased ruminal NDF pool size, which became a greater limitation to DMI as MY of cows increased (Kammes and Allen, 2012b). Because in vitro NDF digestibility is not related to DMI across forage family (i.e., grasses, legumes), in vitro NDF digestibility should be used to compare within forage type only.

The extent to which ruminal distention limits DMI of lactating cows is related to their energy demands determined primarily by MY (Oba and Allen, 1999a; Voelker et al., 2002); DMI is also affected by factors unrelated to the filling effect of rations, as discussed below.

METABOLIC CONTROL

Conrad et al. (1964) proposed that as diet digestibility increases, there is a breakpoint in digestibility at which limitation of DMI by physical fill in the gastrointestinal tract is replaced by satisfaction of energy demand. The "energostatic" control of food intake proposed by Booth (1972) suggested that animals eat to balance energy consumed with energy required. Later, Weston (1996) proposed that the strength of hunger signals is related to the magnitude of the energy deficit. However, these theories fail to explain the negative energy balance in the peripartum period when distention is not a factor limiting DMI. The concept that DMI is limited by energy supplied in excess of energy demand has some merit, but the mechanism is complex and only loosely controlled when based on energy per se.

Mayer (1953) proposed the glucostatic theory of the control of food intake based on research with nonruminant species. However, glucose has little effect on DMI of ruminants; glucose infused intravenously in cows, intraperitoneally in

heifers, intracerebroventricularly in calves, abomasally in lactating cows, and intraportally in sheep had no effect on DMI (Allen, 2000). However, various fuels derived from the fermentation and digestion of feeds can affect DMI of ruminants. Hypophagic effects of propionate infusions have been documented extensively for ruminants (Allen, 2000). Propionate is more hypophagic than acetate or butyrate when infused into the portal vein of sheep (Anil and Forbes, 1980) or mesenteric vein of steers (Elliot et al., 1985). Although propionate might be expected to decrease DMI compared with acetate because it has a higher energy content, propionate linearly decreased total metabolizable energy (ME) intake (diet plus infusates) compared with acetate in lactating cows when infused intraruminally as iso-osmotic mixtures (Oba and Allen, 2003d). These studies suggest that animals probably do not consume feed to meet their energy requirements per se but rather have fuel-specific mechanisms affecting satiety and hunger.

Signal from the Liver

Russek (1963) introduced the idea that the liver is involved in the control of food intake. The liver is in a unique position to monitor changes in fuel metabolism to control eating behavior because of its central role in energy metabolism of animals (Friedman and Stricker, 1976). It is likely involved in regulation of intake because hepatic vagotomy eliminated hypophagic effects of propionate infusion in sheep (Anil and Forbes, 1988) and hypophagic effects of fatty acid (FA) oxidation in rats (Scharrer, 1999). Research with nonruminants suggests that meals can be terminated by a signal carried from the liver to the brain via afferents in the vagus nerve that are affected by hepatic oxidation of fuels and generation of adenosine triphosphate (ATP; Langhans and Scharrer, 1992; Friedman, 1995). The liver is likely to be the primary sensor of energy status because it is a key anabolic organ with the unique advantage of sensing energy supply relative to energy demand (Allen and Bradford, 2012). The signal from the liver can be both stimulatory and inhibitory, depending on the firing rate of vagal afferents; firing rate is increased as energy status decreases, stimulating feeding, and is decreased when energy status increases, inhibiting feeding (Friedman, 1997). Energy requirements of the liver vary over the long term of weeks to months to meet needs for growth and lactation. However, hepatic oxidation of fuels and production of ATP can vary greatly over minutes, affecting feeding behavior and DMI. Normal fluctuations in liver energy status within days might stimulate both hunger and satiety depending on the balance between energy production and utilization. Allen et al. (2009) applied this metabolic control mechanism to ruminant animals and called it the hepatic oxidation theory (HOT) of the control of DMI. Because the liver utilizes fuels from the diet as well as those mobilized from tissues, the control of DMI and energy partitioning are inextricably linked (Allen, 2014).

Acetyl-CoA is the metabolic crossroad that all fuels must be converted to for complete oxidation. However, some fuels are also anaplerotic and can stimulate oxidation of acetyl-CoA in the tricarboxylic acid (TCA) cycle. Whereas some are obligatorily anaplerotic (e.g., propionate, lactate, glutamate), alternate pathways exist for others (e.g., glycerol). Propionate flux to the liver increases greatly during meals (Benson et al., 2002) and is likely a primary satiety signal. Whereas propionate is extensively metabolized by the ruminant liver, there is little net metabolism of acetate (Reynolds, 1995). Ruminant liver has high activity of propionyl-CoA synthetase but not acetyl-CoA synthetase (Ricks and Cook, 1981; Demigne et al., 1986) necessary for activation and subsequent metabolism, thus explaining differences in hypophagic effects of infusions of propionate and acetate in ruminants. Glycerol, like propionate, is a 3-carbon glucose precursor but is less likely to enter the TCA cycle and stimulate oxidation because it enters the gluconeogenic pathway at glyceraldehyde-3-phosphate. Abomasal infusions into lactating cows of propionic acid reduced DMI compared with glycerol infusions (Gualdron-Duarte and Allen, 2017).

Carbohydrates

Cereal grains that are highly fermentable in the rumen can depress DMI of lactating cows; DMI is reduced when less fermentable grains are replaced with more fermentable grain (Allen, 2000; Oba and Allen, 2003b). Depression of DMI by highly fermentable starch sources likely occurs because greater ruminal propionate flux to the liver stimulates hepatic oxidation and generation of ATP, reducing meal size. In contrast, when site of starch digestion is shifted postruminally, a positive response in DMI is likely. This is because the fuels absorbed do not stimulate hepatic oxidation to the same degree as propionate and because the transit time from the rumen to the intestines significantly delays fuel absorption. Starch escaping the rumen is digested to glucose, which is absorbed and partially metabolized to lactate. DMI was reduced by both propionic acid and lactic acid treatments but not by glucose, and only propionic acid decreased ME intake when isocaloric solutions were infused to the abomasum of cows in the postpartum period (Gualdron-Duarte and Allen, 2018). Whereas both propionate and lactate are anaplerotic, liver uptake of lactate is lower than propionate (Reynolds et al., 2003), which might explain different hypophagic effects.

Sugars are highly fermentable in the rumen, but hypophagic effects have not been reported, and they often increase DMI (Oba, 2011). Sugars are generally fermented to butyric acid, which is preferentially oxidized by ruminal epithelia (Weigand et al., 1975). The pectin content of feeds is low, and its effect on DMI is unknown. Whereas pectin is highly fermented in the rumen, most strains of pectin degrading rumen bacteria produce acetic and formic acids and relatively little propionic acid (Dehority, 1969).

Fat

Three review papers were available to examine the effect of various fat sources on DMI (Allen, 2000; Rabiee et al., 2012; Weld and Armentano, 2017). These partially overlapping reviews are consistent in demonstrating that supplemental fats can decrease DMI, but the effects vary by source. A meta-analysis by Weld and Armentano (2017) indicated that saturated fats containing long-chain FAs (LCFAs) tended to increase daily DMI (0.7 kg/3 percent added fat), whereas most other fats significantly decreased DMI (-0.8 to -1.3 kg/3 percent added fat). Exceptions were saturated fats with high palmitic acid content, which did not affect DMI, and medium-chain FAs (MCFAs), which decreased DMI 3.5 kg/3 percent added fat. Increased DMI by saturated fats might be from decreased propionic acid if fats are substituted for cereal grains in rations. Greater decrease in DMI for unsaturated compared with saturated fats is consistent with previous meta-analyses (Allen, 2000; Rabiee et al., 2012). Infusion of FAs into the duodenum depresses DMI if oleic, linoleic, and linolenic acids are present in the mixture but not when palmitic and stearic acid predominate (Drackley et al., 1992); therefore, the effects of feeding unsaturated FA may be explained by postruminal effects (Allen, 2000). Intake depression by unsaturated FA might be related to greater release of cholecystokinin (CCK; Allen, 2000), greater extraction and oxidation in the liver (Allen, 2000), or greater oxidation in enterocytes (Langhans, 2008) compared with saturated FA. The gut peptide CCK may affect satiety centrally or by decreasing rumen emptying, increasing distention. Unsaturated FA increased plasma CCK concentration and decreased DMI compared with saturated FA (Relling and Reynolds, 2007; Bradford et al., 2008), and a CCK receptor antagonist prevented intake depression by a diet high in unsaturated FA (Choi et al., 2000). The large reduction in DMI by MCFAs go directly to the liver from the portal-drained viscera, whereas LCFAs are absorbed in the lymphatic system, and unlike LCFAs, MCFAs do not require protein-mediated transport to cross the mitochondrial membrane (Papamandjaris et al., 1998). However, MCFAs also reduce NDF digestion and could enhance rumen fill effects to limit DMI as well.

Protein

Effects of dietary crude protein (CP) concentrations on DMI are variable and likely from a combination of physical and metabolic mechanisms (Allen, 2000). Increased ration CP increased daily DMI an average of 0.6 kg per percentage unit of CP in 7 of 25 comparisons reported in the literature (Allen, 2000). Roffler et al. (1986) evaluated treatment means from the literature and reported that the marginal response in DMI declined as the CP content of the rations increased. Positive responses in DMI from increased ration CP content is likely partly due to effects of increased rumen

degraded protein effects on ruminal digestibility of feeds (Oldham, 1984) and a reduction in distention as NDF and dry matter (DM) digestibility increase. Although the marginal response in diet DM digestibility from increased ration CP content decreased as the CP content of the ration exceeded 15 percent, it remained positive when ration CP content exceeded 20 percent (Oldham, 1984). Improvements in digestibility of NDF and DM are likely to have greater effects for cows in which control of DMI is dominated by distention. However, positive effects of ration CP content on DMI might also occur when DMI is limited by hepatic oxidation when protein is substituted for starch (Allen, 2000). In addition, if higher CP increases MY this will increase clearance of metabolic fuels from the blood, delaying hepatic oxidation, potentially increasing DMI (Allen, 2014).

Higher dietary CP can increase DMI; however, excessive nonprotein nitrogen can reduce DMI (Conrad et al., 1977). In addition, excess amino acid (AA) and AA imbalances will increase deamination of AA as well as anaplerosis, stimulating oxidation of acetyl-CoA, potentially suppressing DMI (Allen, 2014). Supporting this, a meta-analysis showed that abomasal infusions of casein reduced DMI of cows when supply of metabolizable protein (MP) balance was positive and increased DMI of cows when MP balance was negative (Martineau et al., 2016). Excess AA (when MP supply was positive) likely resulted in greater deamination of AA, stimulating hepatic oxidation and satiety, which was much less likely when MP supply was negative. Besides increased anaplerosis from the AA carbon skeletons, the ammonia from deamination requires detoxification by urea synthesis, which generates an additional carbon skeleton, further contributing to oxidative metabolism in the liver (Oba and Allen, 2003e). Consistent with this, ammonium, compared with sodium, increased the hypophagic effects of propionate when infused intraruminally in lactating cows (Oba and Allen, 2003e). Meta-analyses have shown that supplementation of rumen-protected methionine has had variable effects on DMI dependent on its chemical form and the main forage source of the diet (Patton, 2010; Zanton et al., 2014). Supplying limiting AA will likely decrease anaplerosis and potentially increase DMI depending on the next most limiting AA and the extent to which hepatic oxidation is limiting DMI (Allen, 2014).

Interaction with Physiological State

Feeding behavior response to diet is also affected by physiological state. Variation in the hypophagic effects of propionate might be related to the balance between flux of propionate to the liver and rate of utilization of propionate for gluconeogenesis, affected by glucose (i.e., lactose synthesis) demand (Allen, 2000). The maximum rate of gluconeogenesis at any point in time is regulated by hormones such as insulin and glucagon, and increased glucose demand is expected to increase gluconeogenic capacity (McGrane, 2000). According to the HOT, when glucose demand is

high, gluconeogenesis increases, and this should decrease TCA cycle activity and oxidation of acetyl-CoA, resulting in greater meal size. Conversely, when glucose demand is low, TCA activity is increased, generating ATP, resulting in satiety and smaller meal size. In support of this, the marginal depression in DMI from propionate infusion was positively related to plasma glucose concentration of cows (Oba and Allen, 2003c).

Hyperlipidemia in the periparturient period is common, and uptake of nonesterified FAs (NEFAs) by the liver increases greatly (Reynolds et al., 2003), resulting in increased FA oxidation, buildup of acetyl-CoA, and hepatic export of ketone bodies. Intake might be suppressed by hepatic FA oxidation and generation of ATP. The negative association between body condition score (BCS) at parturition and DMI in the postpartum period reported by most studies (Roche et al., 2009) is likely related to supply of acetyl-CoA. In addition, cows with greater fat mobilization during the postpartum period had a greater depression in DMI and greater negative energy balance compared with cows with less fat mobilization (Weber et al., 2013).

The primary source of acetyl-CoA is from mitochondrial β -oxidation of NEFA, but all fuels (lactate, glycerol, AA) that are completely oxidized in the mitochondria must enter the TCA cycle via metabolism to acetyl-CoA. Hypophagic effects of anaplerotic metabolites appear to be dependent on the availability of acetyl-CoA to be oxidized (Stocks and Allen, 2012, 2013). Consistent with this, lower doses of propionate were more hypophagic, reducing total ME intake for cows in the postpartum period that were in a lipolytic state compared with cows in mid-lactation (Oba and Allen, 2003a). The concentration of acetyl-CoA in hepatocytes varies with changes in physiological state over the long term as well as diurnally (Piantoni et al., 2015). The postprandial reduction in NEFA supply to the liver, affected by insulin secretion and sensitivity of adipose tissue, likely affects feeding behavior by reducing acetyl-CoA available for oxidation. Following the initiation of meals, plasma insulin concentration increases and plasma NEFA (Allen et al., 2005) and liver acetyl-CoA concentrations decrease (Piantoni et al., 2015). DMI by cows in the postprandial period was positively related with the extent to which plasma NEFA and hepatic acetyl-CoA concentrations decreased following the initiation of meals following feeding (Piantoni et al., 2015).

Cows in the postpartum period require glucose precursors, primarily from dietary starch, to satisfy glucose demand from the rapid increase in MY while DMI is depressed. However, ruminal fermentation of starch results in increased propionate production that can result in satiety. Several experiments have fed rations differing in starch content in the postpartum period (Andersen et al., 2003; Rabelo et al., 2005; Dann and Nelson, 2011; McCarthy et al., 2015; Alborno and Allen, 2018). Increasing ration starch content increased DMI and MY in experiments reported by Andersen et al. (2003) and

Rabelo et al. (2005), but in those experiments, cereal grains were substituted for forage, decreasing the forage NDF content of the ration and the contribution of ruminal distention to satiety, especially as lactation progressed and the lipolytic state diminished. Substitution of corn grain for NFFS that have much less filling effect than forage NDF had different effects in other experiments. Dann and Nelson (2011) substituted corn meal for NFFS to increase ration starch content from 21 percent to 25.5 percent, and the higher starch diet decreased DMI 1.5 kg/d, whereas McCarthy et al. (2015) substituted ground corn for NFFS to increase ration starch content from 21.5 percent to 26.2 percent, and the higher starch diet had no effect on DMI (kg/d) overall but interacted with time; treatment did not affect DMI during the first 2 weeks postpartum, but the higher starch diet began to increase DMI slightly by the third week postpartum. Alborno and Allen (2018) reported that high moisture corn reduced DMI compared with dry ground corn with a greater reduction when fed in 28 percent starch rations compared with 22 percent starch rations. Increased ruminal fermentability of starch also decreased DMI of cows in the postpartum period when steam-flaked corn was substituted for cracked corn (Dann et al., 1999) and when barley was substituted for corn (Sadri et al., 2009) but did not affect DMI when high moisture corn was substituted for dry ground corn in a high (27 percent) forage NDF ration (Rockwell and Allen, 2016). It is likely that DMI by cows in the postpartum period is controlled by hepatic oxidation as well as ruminal distention and that the dominant mechanism is dependent on the availability of acetyl-CoA for oxidation, ruminal starch fermentability, and the filling effect of the diet. Feeding a moderately fermentable starch source (e.g., dry ground corn) will likely allow higher starch rations to be fed, restoring euglycemia sooner.

Decreased DMI from excessive ruminal fermentation might also be caused by an inflammatory response to lipopolysaccharide released from ruminal microbial lysis (Bradford et al., 2015). Insulin resistance is induced by inflammatory signals contributing to the lipolytic state (Bradford et al., 2015). The reduction in DMI might be from central effects of acute-phase proteins (Sartin et al., 2011) as well as increased anaplerosis from propionate and increased availability of acetyl-CoA for hepatic oxidation.

INTEGRATION OF SIGNALS

The signal from the liver and other peripheral signals are relayed to brain-feeding centers by sensory nerves and integrated to affect feeding behavior. The relative contribution of different signals likely varies temporally, within and across days. Intake is likely limited by distention of undigested feed residues in the gastrointestinal tract when energy requirements are high and by mechanisms related to specific fuels from the diet or body reserves in the immediate postpartum period and as MY declines through lactation. Signals related

to distension and metabolism have synergistic effects on DMI by ruminants. Additive effects have been reported for intraruminal infusion of short-chain FAs on DMI of lactating cows when rumens were distended by balloons (Mbanya et al., 1993) and forage NDF concentration of the diet (Choi and Allen, 1999). Therefore, mechanisms are not mutually exclusive, and different mechanisms can both contribute to, or dominate, control of feeding within days.

Homeorhetic mechanisms that affect DMI and energy partitioning over the long term of weeks or months, such as those related to insulin, growth hormone, and leptin, influence hepatic oxidation by affecting the supply of NEFA, glycerol, and AA to the liver and therefore short-term control of DMI by hepatic oxidation (Allen, 2014). Whereas many of leptin's effects on DMI are thought to be mediated centrally (Houseknecht et al., 1998), its lipolytic effects increase NEFA available for hepatic oxidation, likely contributing to satiety (Allen, 2014). Leptin concentration in blood was positively related to BCS of lactating cows (Bradford et al., 2006), and its effects on satiety are consistent with the lipostatic theory of body weight (BW) maintenance (Kennedy, 1953). Therefore, mechanisms controlling energy intake and partitioning are entwined and inseparable and are affected by both diet and physiological state of cows (Allen, 2020). The dominant mechanism affecting DMI will vary among cows fed the same ration, affecting their responses to changes in ration composition and providing an opportunity to optimize feeding strategies for cows fed individually or in groups (Allen and Piantoni, 2014).

Consistently accurate prediction of DMI in ruminants has been difficult to achieve because of an incomplete understanding and availability of data to describe the interactions among diet composition, physiological state, environment, and management factors affecting DMI. For additional discussions and reviews on intake, see Bade and McLaughlin (1987), NRC (1987), Forbes (2007), Roche et al. (2008), and Allen (2014, 2020).

EQUATIONS FOR PREDICTING DRY MATTER INTAKE

Lactating Cows

In lactating dairy cattle, milk production (energy expenditure) usually peaks 4 to 8 weeks postpartum, and peak DMI (energy intake) lags until about 10 weeks postpartum (NRC, 1989). Earlier editions of Nutrient Requirements of Dairy Cattle used various approaches to predict DMI. The last edition of this report (NRC, 2001) included an empirical equation to estimate DMI of lactating Holstein cows with the inclusion of only animal factors that could be easily measured or known. Several equations were evaluated using weekly data from both published and unpublished studies. The best overall prediction equation was the equation of Rayburn and Fox (1993), multiplied by an adjustment for week of lactation to account for depressed DMI during early

lactation developed by Roseler et al. (1997). Ration composition was not included because the model was evaluative, and it was assumed cows were consuming feed relative to their energy requirements (except in early lactation). Therefore, the estimated DMI represents the mean effects of diets used in the studies and their interaction with animal factors. No published DMI data were available for developing or modifying the equation for use with breeds other than Holstein, and readers were referred to Holter et al. (1996) for the prediction of DMI of Jersey cattle. The DMI prediction equations (NRC, 2001) for primiparous and multiparous cows were similar, so no adjustment was made for parity per se (BW and MY will account for most parity effects).

An empirical equation (Equation 2-1) derived by de Souza et al. (2019) from a data set composed of 31,635 weekly observations for 3,143 lactations (1,462 primiparous and 1,681 multiparous) on 2,791 cows (de Souza et al., 2019) is used to estimate DMI in the model. Data were collected from cows at 11 research stations across the United States from 2007 to 2016. The data set includes animal factors only for cows between 1 and 368 days in milk (DIM) fed a range of rations with mean $\pm$ standard deviation (SD) of 105 $\pm$ 50 days postpartum, 24.3 $\pm$ 4.55 kg DMI, 29.9 $\pm$ 6.23 Mcal/d milk net energy (MilkE), 624 $\pm$ 80.2 kg BW, 3.03 $\pm$ 0.459 BCS, 0.021 $\pm$ 1.22 kg/d BW change, and 149 $\pm$ 5.28 cm height. All cows were housed in confinement and fed a total mixed ration (TMR) once per day and milked two or three times per day. To predict DMI for lactating Holstein cows, Equation 2-1 includes MilkE, BW, BCS, an adjustment for parity, and a nonlinear adjustment to account for depressed DMI in early lactation:

$$\begin{aligned} \text{DMI (kg/d)} = & [3.7 + \text{Parity} \times 5.7] + 0.305 \\ & \times \text{MilkE (Mcal/d)} + 0.022 \times \text{BW (kg)} \\ & + (-0.689 - 1.87 \times \text{Parity}) \times \text{BCS} \times [1 - (0.212 \\ & + \text{Parity} \times 0.136) \times e^{(-0.053 \times \text{DIM})}] \end{aligned}$$

(Equation 2-1)

where parity is an adjustment factor ranging from 0 (all primiparous) to 1 (all multiparous) and BCS is scaled from 1 (thin) to 5 (obese). The equation predicted DMI with a small mean bias and high accuracy and precision; the fit statistics for the cross-validation across studies of the equation using the modeling data set were root mean squared error of prediction (RMSEP) = 2.61 kg, mean bias = 0.008 kg, mean bias, percent of mean squared error (MSE) = 6.9, slope bias, % MSE = 2.28, and concordance correlation coefficient = 0.80.

Equation 2-1 was validated against the equation recommended in the previous report (NRC, 2001) using an independent data set comprising 9,050 weekly observations (de Souza et al., 2019). Equation 2-1 was superior for predicting DMI compared with the equation recommended in the previous report, especially for cows past peak lactation. Improved prediction of DMI in mid- to late lactation is likely because parity, BCS, and their interaction were included in Equations

tion 2-1; as BCS increases, DMI is reduced by 0.70 and 2.6 kg/d per unit of BCS for primiparous and multiparous cows, respectively. In addition, the adjustment for the depression in DMI in early lactation was affected by parity in Equation 2-1 with a greater reduction for multiparous compared with primiparous cows. Compared with the previous equation (NRC, 2001), Equation 2-1 has a lower DMI per unit of milk energy output and a similar coefficient for the energy required for maintenance (de Souza et al., 2019). The lower DMI per unit of milk energy output is likely because of improvements in production and feeding over the two decades between the data sets used to develop the equations.

Equation 2-1 was developed using data exclusively from Holstein cows and is most appropriate to predict DMI for Holsteins. However, the equation predicted DMI reasonably well for Jersey and Holstein crossbred cows in a limited data set of treatment means reported in the literature (de Souza et al., 2019).

Ration Effects

Equation 2-1 predicts DMI of cows consuming rations with a wide range of composition. Accuracy of DMI prediction is limited by the potentially large effects of ration composition on DMI that are not accounted for when using animal factors only. Ration factors include those that affect DMI by their filling effects through distention as well as by metabolic effects. Including these factors should improve accuracy of DMI prediction, but accuracy is ultimately limited because of the many interactions among ration factors and physiological state. In addition, diet can limit or stimulate MY that can then affect DMI, increasing the complexity of DMI prediction. However, limited recommendations can be made to adjust predicted DMI for some ration characteristics.

This report includes an equation that combines factors related to the filling effect of rations and MY to help assess the effects of ration composition on DMI of lactating cows during ration formulation (Allen et al., 2019). The data set used includes 134 treatment means from 34 experiments reported in 32 peer-reviewed articles published from 1990 through 2015. It includes data for cows ranging from 60 to 309 days postpartum with a mean $\pm$ SD of 107 $\pm$ 48 days postpartum, 23.0 + 2.8 kg DMI, 32.0 + 7.5 kg/d MY within study, 643 $\pm$ 59 kg BW, and for ration concentrations (percent of DM) of 17.8 $\pm$ 1.6 for CP, 34.1 $\pm$ 4.6 for NDF, 20.5 $\pm$ 4.0 for acid detergent fiber (ADF), and 23.9 $\pm$ 5.7 for forage NDF (fNDF), as well as ration ADF/NDF of 0.600 $\pm$ 0.083 and a laboratory measure of NDF digestibility (in vitro or in situ, fNDFD) for the sole forage or major forage of 52.0 $\pm$ 12.3. The ratio of ADF to NDF was included as a proxy for forage fragility (Allen and Piantoni, 2014); legumes, with an ADF/NDF of $\sim$ 0.8, are more susceptible to comminution than perennial grasses with an ADF/NDF typically -0.6 or less (Voelker-Linton and Allen, 2008; Kammes

and Allen, 2012a). The full model included the linear and quadratic effects of ration CP, ADF, NDF, fNDF, ADF/NDF, and fNDFD, as well as the linear and quadratic interactions among ration factors, BW, and the mean MY for each study and its interaction with the ration factors. The mean MY for each study was included because distention likely becomes a more dominant mechanism limiting DMI as MY increases (Oba and Allen, 1999a; Voelker et al., 2002). The full model was reduced by backward stepwise regression to select the model with the lowest Bayesian information criterion and evaluated using a 5-fold cross-validation. The final equation was

$$\begin{aligned} \text{DMI (kg/d)} = & 12.0 - 0.107 \times \text{fNDF} + 8.17 \times \text{ADF/NDF} \\ & + 0.0253 \times \text{fNDFD} - 0.328 \times (\text{ADF/NDF} - 0.602) \\ & \times (\text{fNDFD} - 48.3) + 0.225 \times \text{MY} + 0.00390 \\ & \times (\text{fNDFD} - 48.3) \times (\text{MY} - 33.1) \end{aligned} \quad (\text{Equation 2-2})$$

where DMI = dry matter intake, kg/d; fNDF = forage NDF content of diet, percentage; ADF/NDF = ADF as a fraction of NDF in the diet; fNDFD = digestibility of forage NDF measured in vitro or in situ, percentage; and MY = milk yield, kg/d. The observed DMI versus predicted DMI had an intercept of -0.05 kg and a slope close to unity (1.006), with a mean bias of 0.00 kg/d, a root mean square error (RMSE) of 1.55, and concordance correlation coefficient (CCC) of 0.83 (Allen et al., 2019).

DMI was positively related with MY and ADF/NDF and negatively related with fNDF. DMI and fNDFD were related positively for cows with high MY but related negatively for cows with low MY, diminishing the overall effect of fNDFD on DMI. In addition, DMI increased with MY to a greater extent for high fNDFD compared with low fNDFD. DMI and fNDFD were related positively for low-ration ADF/NDF but related negatively for high-ration ADF/NDF. Also, DMI and ration ADF/NDF were related positively when fNDFD was low, but DMI was not affected by ration ADF/NDF when fNDFD was high.

The primary equation to predict DMI in the present report includes animal factors only (Equation 2-1). Equation 2-2 is included to evaluate DMI once feeds and their dietary proportions are determined within the limitations mentioned by Allen et al. (2019). The equation should be limited to cows past 60 days postpartum because data from cows earlier in lactation were not included in the data set and because the control of DMI is likely dominated by metabolic mechanisms rather than distention for cows in the postpartum period. The data set included data from Holstein cows only. Because BW was not significant and not included in the final equation, the suitability of the equation for smaller breeds such as Jerseys and Holstein crossbreds is unknown. The data set did not include treatments with NFFS or ground or finely chopped forages. Whereas inclusion of NFFS in rations has been shown to have little effect on DMI of lactating cows (Allen,

2000), the ADF/NDF of NFFS varies widely, so evaluating effects of large changes in NFFS on DMI is not recommended. In addition, ground, pelleted, or very finely chopped forages should not be classified as forage for the purposes of predicting DMI with this equation. The rations included in the data set contained a single forage, or other forages contributed little to the ration. It is suggested that a weighted average be used for fNDFD when this equation is used for rations with multiple forages or that the mean fNDFD of the data set (52.0 percent) be used if a measure of fNDFD is not available. In addition, because it is a biological assay with expected variation across runs, fNDFD for all forage comparisons should be measured with the same method and incubation time and preferably within the same run or corrected using the same laboratory standard.

Besides adjusting for the filling effect of rations, fat sources containing unsaturated fats (e.g., oil, tallow, calcium salts palm, calcium salts LCFA) depressed DMI with a mean reduction of -0.41 kg/1 percent added fat (Weld and Armenitano, 2017). Whereas DMI can be increased by increasing the protein concentration of rations and reduced by propionic acid produced from ruminal starch fermentation, variation in response by physiological state and lack of sufficient data preclude recommendations to adjust DMI using these inputs. Diets also affect milk energy output and BW gain, so dietary adjustments to DMI should be used with caution.

Growing Heifers

The previous report (NRC, 2001) evaluated equations to predict DMI of growing heifers, including those developed by Stallings et al. (1985) using animal factors only, as well as equations that included both animal and ration factors developed by Quigley et al. (1986) and the calf equation from the Nutrient Requirements of Beef Cattle (NRC, 1996). After an initial evaluation using a small data set, the NRC equation for beef cattle was selected and validated with a larger data set of 2,727 individual observations on growing heifers ranging from 58 to 588 kg. However, validation statistics were not provided, and visual observation of the plot of observed versus predicted DMI indicates that the equation increasingly underpredicts DMI as DMI increases from 2 to 12 kg/d. Anele et al. (2014) reevaluated the equation from the previous edition of the Nutrient Requirements of Beef Cattle (NRC, 1996), with a much larger database of growing and finishing beef cattle and developed alternative equations to predict DMI. Based on the results of that evaluation, the Committee on Nutrient Requirements of Beef Cattle (NASEM, 2016) recommended continued use of the calf equation from the previous report to predict DMI of growing-finishing beef cattle, although they also presented alternative equations.

Hoffman et al. (2008) evaluated the equation recommended in the previous edition (NRC, 2001) and the equation developed by Quigley et al. (1986) using a database of daily pen DMI from 44 pens of Holsteins and 30 pens of

crossbred heifers collected over 28 months (Maltecca et al., 2006). They determined that both equations were reasonably accurate but had significant DMI $\times$ BW prediction bias, over- or underpredicting DMI of light or heavy heifers. They developed alternative exponential or mixed models using BW only, or BW and ration NDF content, as well as equations using BW, ration NDF content, and temperature for Holstein and for crossbred heifers. The exponential model with a single BW (kg) term has similar precision compared with the equations reported by NRC (2001) and Quigley et al. (1986) but decreased DMI prediction bias. The equation for Holsteins from Hoffman et al. (2008) was modified to include mature BW (MatBW) so that it could be applied to all breeds (Equation 2-3). Holsteins were assumed to have a MatBW of 700 kg:

$$\text{DMI, kg/d} = 0.022 \times \text{MatBW} \times (1 - e^{-1.54 \times (\text{BW}/\text{MatBW})}) \quad (\text{Equation 2-3})$$

where MatBW is the expected MatBW of the heifer.

When Hoffman et al. (2008) included ration NDF in the DMI equation, it modestly increased R^2 , decreased standard errors of prediction, and eliminated bias compared to the equation that only used BW. The Holstein-based NDF equation was modified to include MatBW so that it could be applied to all breeds (Equation 2-4):

$$\text{DMI, kg/d} = [0.0226 \times \text{MatBW} \times (1 - \exp\{-1.47 \times (\text{BW}/\text{MatBW})\})] - [0.082 \times (\text{NDF} - \{23.1 + 56 \times (\text{BW}/\text{MatBW}) - 30.6 \times (\text{BW}/\text{MatBW})^2\})] \quad (\text{Equation 2-4})$$

where NDF is the NDF concentration (DM basis) of the diet.

Whereas the temperature effect on DMI was significant, its effect was small and prediction bias was observed for both light and heavy heifers. Equation 2-3 is recommended to predict DMI for heifers when diet NDF concentrations are not known. Equation 2-4 is recommended to estimate DMI when ration NDF is known with confidence. However, because of the diversity in diets, environments, and numerous other factors, users are encouraged to measure actual DMI rather than relying on estimated values.

FEEDING MANAGEMENT, FEEDING BEHAVIOR, AND ENVIRONMENTAL FACTORS AFFECTING FEED INTAKE

The goal of any feeding system is to provide the opportunity for cows to consume the amount of feed specified in a formulated ration. Considerations in choosing a feeding system should include housing facilities, equipment necessities, herd size, labor availability, and cost. Nutrients can be effectively supplied by feeding either a TMR or individual ingredients. A TMR allows for the mixing of all feed ingredients together based on a prescribed amount of

each ingredient. When consumed as a TMR without sorting of ingredients, more even rumen fermentation and a better use of nutrients should occur than when feeding ingredients separately. Electronic feeders reduce the labor involved in individual-concentrate feeding and provide an opportunity to feed concentrates through several feedings of smaller amounts each day. Feeding forages and concentrates separately limit the accuracy of ration formulation when forages are provided free choice and the amount fed is usually unknown or individual cow amounts are calculated from a group average intake. A partially mixed ration provides some of the advantages of a TMR while allowing concentrates to be fed according to the needs of individual cows.

Eating Habits and Cow Behavior

Early lactation cows (63 DIM) producing 23 to 44 kg of milk per day fed a TMR ad libitum ate an average of 5 hours per day (Dado and Allen, 1994). Cows in that study were housed in tie-stalls, were fed twice per day, and had access to feed 22 hours per day. Meal frequency ranged from 9 to 13 (mean of 11) eating bouts per day that averaged 29 minutes per bout. Mean DMI at each eating bout was about 10 percent of the total daily DMI, which ranged from 15 to 27 kg/d. However, meal size was highly variable within a day, with larger meals following feeding and smaller meals at night. DMI and MY data from multiple studies were positively related to eating time and ruminating time per day as well as meal frequency (Johnson and DeVries, 2018). Eating time can be affected by over an hour per day by differences in concentration of forage NDF, forage NDF digestibility, and particle size, which might limit DMI under competitive feeding situations (Grant and Ferraretto, 2018).

Behavior at the feed bunk is often affected by social dominance. Dominant cows, usually older and larger, tend to spend more time eating than do cows with a lower social rank in a competitive situation, such as when bunk space is restricted (Albright, 1993). Socially dominant animals, not necessarily the highest producers, tend to consume more feed at the bunk in these situations (Friend and Polan, 1974). Competition at the feed bunk increased meal size and length and decreased meal frequency, with no effect on DMI or total eating time in one study (Hosseinkhani et al., 2008), but reduced eating time and increased eating rate, with a slight increase in DMI in another study (Olofsson, 1999). Martinsen (1992) and Martinsson and Burstedt (1990) found that limiting the access to feed to 8 hours per day decreased milk production of cows averaging about 25 kg/d by 5 to 7 percent compared with cows that had free-choice access to feed.

Albright (1993) recommended at least 46 cm of bunk space per cow. However, the optimal or critical feed bunk space needed probably varies depending on competition among cows, the total number of cows having access to the feed space, and the availability of feed over a 24-hour period. Friend et al. (1977) evaluated bunk spaces of 50, 40, 30, 20,

and 10 cm per cow, for early lactation cows with mature equivalent productions of 7,700 to 10,000 kg/y. Average time spent at the feed bunk (3.7 hours/d) did not decrease until only 10 cm of space per cow was available. When there was 20 or 10 cm per cow, the correlation of dominance to duration of eating periods increased. For growing dairy heifers, feed-bunk space requirement varies with age. Longenbach et al. (1999) found that rapid growth in growing heifers fed a TMR could be maintained in young heifers (4 to 8 months old) with 15 cm of bunk space. But, by the age of 17 to 21 months, feed bunk space needed to be similar (47 cm) to that recommended for lactating cows.

Cattle prefer mangers that allow them to eat off a smooth surface in a natural grazing position. Albright (1993) cited evidence showing cows eating with their heads down produce 17 percent more saliva than cows eating with their heads in a horizontal position. Feed-wasting activities associated with elevated bunks, such as feed tossing, are eliminated when cows eat with their heads down (Albright, 1993).

Feeding Frequency

Whereas several studies have reported effects of feeding frequency on milk production and on feeding and sorting behavior, few studies have reported effects of feeding frequency on DMI. Potential benefits of increased feeding frequency might be from metabolic responses from a more consistent ruminal fermentation and supply of absorbed fuels. However, Hart et al. (2014) reported that frequency of feed delivery had no effect on DMI by primiparous cows and little effect on DMI by multiparous cows (30.6, 29.7, and 31.1 kg/d for one, two, and three times per day feeding, respectively) and did not affect MY for either group when housed in free stalls. In addition, DMI and MY were not affected by feeding twice per day compared with feeding once per day (Niu et al., 2014), and increasing frequency of concentrate feeding from two or six times per day had no effect on DMI, yield of milk and milk components, or ruminal fermentation characteristics (Macleod et al., 1994). Increased frequency of pushing up feed to cows did not affect lying time, yield of milk and milk components, or feed sorting for cows housed in tie-stalls (Miller Cushon and DeVries, 2017) and had little effect on feeding time or diurnal pattern of feed alley attendance for cows in free stalls (DeVries et al., 2003). Delivering feed once per day to cows ad libitum in the morning (0830 h) compared with the evening (2030 h) affected daily rhythms of feeding and lying behavior in both the winter (Niu et al., 2014) and summer months in Pennsylvania (Niu and Harvatine, 2018) but did not alter MY or composition in either study or DMI during the winter months. However, delivering feed in the evening decreased DMI and digestibility of DM and NDF compared with delivering feed in the morning during the summer months. Delivering feed to cows between milkings (twice per day) tended to decrease DMI compared with delivery at milking time ($P < 0.10$),

27.2 versus 26.5 kg/d) with no effect of milk or milk components, increasing efficiency of milk production (4 percent fat-corrected milk/DMI, 1.83 versus 1.67; King et al., 2016). Cows ate smaller meals more slowly and more frequently, possibly promoting a more stable rumen environment.

Sorting

Sorting of rations is affected by frequency of feeding and ration characteristics and varies among cows. Sorting was reduced by increasing frequency of feeding from once to twice per day (DeVries et al., 2005) and by adding a molasses-based liquid feed (DeVries and Gill, 2012).

Supplementing a molasses-based liquid feed in high-straw, dry cow diets improved DMI and reduced sorting against long particles but increased sorting against short particles (Havekes et al., 2020a). Addition of water to decrease the DM concentration of a ration from 81 percent to 64 percent reduced sorting and tended to increase NDF intake and milk fat percentage but did not affect DMI or MY (Leonardi et al., 2005). In addition, adding water to a high-straw diet improved DMI and reduced sorting against the longest ration particles by cows in the dry period (Havekes et al., 2020c). However, addition of water during summer months to decrease the DM concentration of a ration from 56 percent to 51 percent or 44 percent increased sorting and the temperature of feed in the hours after feeding and decreased DMI, suggesting that adding water during high ambient temperatures might have accelerated spoilage (Felton and DeVries, 2010). Feeding shorter chopped wheat straw in a high-straw diet improved DMI and reduced sorting by cows in the dry period (Havekes et al., 2020b). Adding long hay to rations increased selective consumption of fine particles but also increased intake of longer particles because of their greater concentration in the ration, but quality of hay had no effect on sorting (Leonardi and Armentano, 2003). Whereas cows consistently sorted against long particles in favor of fine particles, sorting was highly variable among cows (Leonardi and Armentano, 2003). Primiparous cows sorted more than multiparous cows, but sorting was not affected by stage of lactation (DeVries et al., 2011).

Grouping Cows

Grouping cows according to their nutrient requirements can help optimize MY and efficiency of milk production. Mechanisms controlling DMI and energy partitioning vary across a lactation, and ration composition should be altered to maximize energy intake in early lactation and partition energy to MY and limit gain in body condition later in lactation (Allen and Piantoni, 2014). Fresh cows should be grouped separately and offered a more filling ration with moderately fermentable starch sources until distention begins to limit DMI. The more filling ration will likely reduce risk of displaced abomasum and help maintain rumen pH by

buffering rumen contents. However, control of DMI by cows in early to mid-lactation is dominated by ruminal distension, and they benefit from rations that are less filling with higher starch contents to achieve and maintain high MY. Cows in late lactation will likely gain excessive body condition if offered this ration, which might compromise health when the cows calve. Therefore, once cows replete body reserves past peak lactation, they should be offered a ration with less starch content to maintain body condition and MY (Allen and Piantoni, 2014). Primiparous cows do not peak in DMI as early in lactation, but they are more persistent in DMI after peak than are multiparous cows. If possible, primiparous and multiparous cows should be grouped separately because of differences in DMI and social hierarchy. Primiparous cows are usually more timid and of lower social rank in the herd initially, but they gradually rise in social rank as more cows enter the herd or as older cows leave (Wierenga, 1990). Phelps and Drew (1992) reported an increase of 725 kg in milk over a 305-day lactation for first-lactation animals when grouped separately instead of being mixed in with older cows. However, although primiparous cows grouped alone visited the robotic milking unit and feed troughs more frequently, they spent less time eating with no effect on DMI or MY than when housed in a group of 30 percent primiparous and 70 percent multiparous cows (Bach et al., 2005). Production response to grouping primiparous cows separately likely varies depending on group size and bunk space per cow.

Weather

The thermal neutral zone of dairy cattle is about 5 to 20°C but varies among animals. Temperatures below or above that zone alter intake and metabolic activity. Ruminants adapt to chronic cold stress conditions by increasing thermal insulation, basal metabolism, and DMI (Young, 1983). Rumination activity, reticulo-rumen motility, and rate of passage are also increased (Young, 1983). However, in extreme cold, increased DMI may not compensate for the increased metabolic rate, so animals may shift energy use from productive purposes to heat production. A rise in ambient temperature above the thermal neutral zone decreases DMI and milk production. Acute heat stress with average temperature humidity index (THI) of 72 decreased DMI of lactating cows 11.5 percent compared with when they were in a thermoneutral environment with an average THI of 57 (Collier et al., 2018). However, heat stress can decrease DMI by cows more than 50 percent compared with a thermoneutral environment (NRC, 1981) and the reduction in DMI to acute heat stress increases for cows with higher MY (Collier et al., 2018). Water consumption by cattle increases with ambient temperature, but acute heat stress can decrease water intake along with DMI and milk synthesis, and the reduction in water intake increases for cows with higher MY (Collier et al., 2018).

Collier et al. (1981) suggested that the effects of heat stress on MY had a lag of 24 to 48 hours, and West et al.

(2003) reported that DMI was most sensitive to mean air temperature 2 days before, declining 0.85 kg/d for each degree C increase in mean air temperature over the range of 25 to 32°C, whereas MY was most sensitive to THI, declining 0.88 kg/d for each unit increase in THI over the range in THI of 73 to 83°C for Holstein cows. Furthermore, the declines in MY and DMI were substantially less when evaluated with climatic measures on the same day compared with those measured 2 days earlier. Igono et al. (1992) reported that the diurnal pattern of temperature during the day affected MY and that a cool period of less than 21°C will minimize the decline in MY. The equations used for predicting DMI of lactating cows in this report do not include a temperature or humidity adjustment factor because of insufficient DMI data outside of the thermoneutral zone to validate equation modifiers. During periods of heat stress, actual DMI will likely be greater than predicted from the reduction in MY; the reduction in DMI accounted for only 50 percent of heat stress-induced decrease in MY (Wheelock et al., 2010).

REFERENCES

- Albomoz, R. I., and M. S. Allen. 2018. Highly fermentable starch at different diet starch concentrations decreases feed intake and milk yield of cows in the early postpartum period. *J. Dairy Sci.* 101:8902-8915.
- Albright, J. L. 1993. Feeding behavior of dairy cattle. *J. Dairy Sci.* 76:485-498.
- Allen, M. S. 1996. Physical constraints on voluntary intake of forages by ruminants. *J. Anim. Sci.* 74:3063-3075.
- Allen, M. S. 2000. Effects of diet on short-term regulation of feed intake by lactating dairy cattle. *J. Dairy Sci.* 83:1598-1624.
- Allen, M. S. 2014. Drives and limits to feed intake in ruminants. *Anim. Prod. Sci.* 54:1513-1524.
- Allen, M. S. 2020. Review: Control of feed intake by hepatic oxidation: integration of homeostasis and homeorhesis. *Animal* 14(S 1):s55-s64.
- Allen, M. S., and B. J. Bradford. 2012. Control of food intake by metabolism of fuels—a comparison across species. *Proc. Nutr. Soc.* 71:401A109.
- Allen, M. S., and P. Piantoni. 2014. Carbohydrate nutrition: Managing energy intake and partitioning through lactation. *Vet. Clin. North Am. Food Anim. Pract.* 30:577-597.
- Allen, M. S., B. J. Bradford, and K. J. Harvatine. 2005. The cow as a model to study food intake regulation. *Annual Review of Nutrition* 25:523-547.
- Allen, M. S., B. J. Bradford, and M. Oba. 2009. Board invited review: The hepatic oxidation theory of the control of feed intake and its application to ruminants. *Journal of Animal Science* 87:3317-3334.
- Allen, M. S., D. O. Sousa, and M. J. VandeHaar. 2019. Equation to predict feed intake response by lactating cows to factors related to the filling effect of rations. *J. Dairy Sci.* 102:7961-7969.
- Andersen, J. B., N. C. Friggens, K. Sejrsen, M. T. Sorensen, L. Munksgaard, and K. L. Ingvarsen. 2003. The effects of low vs. high concentrate level in the diet on performance in cows milked two or three times daily in early lactation. *Livestock Prod. Sci.* 81:119-128.
- Anel, U. Y., E. M. Dobby, and M. L. Galyean. 2014. Predicting dry matter intake by growing and finishing beef cattle: Evaluation of current methods and equation development. *J. Anim. Sci.* 92:2660-2667.
- Anil, M. H., and J. M. Forbes. 1980. Feeding in sheep during intraportal infusions of short-chain fatty acids and the effect of liver denervation. *J. Physiol.* 298:407-414.
- Anil, M. H., and J. M. Forbes. 1988. The roles of hepatic nerves in the reduction of food intake as a consequence of intraportal sodium propionate administration in sheep. *Q. J. Exp. Physiol.* 73:539-546.
- Bach, A., C. Iglesias, M. Devant, and N. Rafols. 2005. Performance and feeding behavior of primiparous cows loose housed alone or together with multiparous cows. *J. Dairy Sci.* 89:337-342.
- Baile, C. A., and J. M. Forbes. 1974. Control of feed intake and regulation of energy balance in ruminants. *Physiol. Rev.* 54:161.
- Baile, C. A., and C. L. McLaughlin. 1987. Mechanisms controlling feed intake in ruminants: A review. *J. Anim. Sci.* 64:915-922.
- Balch, C. C., and R. C. Campling. 1962. Regulation of voluntary intake in ruminants. *Nutr. Abstracts Rev.* 32:669.
- Benson, J. A., C. K. Reynolds, P. C. Aikman, B. Lupoli, and D. E. Beever. 2002. Effects of abomasal vegetable oil infusion on splanchnic nutrient metabolism in lactating dairy cows. *J. Dairy Sci.* 85:1804-1814.
- Blaxter, K. L., F. W. Wainman, and R. S. Wilson. 1961. The regulation of food intake by sheep. *Anim. Prod.* 3:51.
- Booth, D. A. 1972. Postabsorptively induced suppression of appetite and the energostatic control of feeding. *Physiol. Behav.* 9:199-202.
- Bradford, B. J., M. Oba, R. Ehrhardt, Y. R. Boisclair, and M. S. Allen. 2006. Propionate is not an important regulator of plasma leptin in dairy cows. *Domestic Anim. Endocrinol.* 30(2):65-75.
- Bradford, B. J., K. J. Harvatine, and M. S. Allen. 2008. Dietary unsaturated fatty acids increase plasma GLP-1 and CCK and may decrease pre-meal ghrelin in lactating cows. *J. Dairy Sci.* 91:1443-1450.
- Bradford, B. J., K. Yuan, J. K. Famey, L. K. Mamedova, and A. J. Carpenter. 2015. Inflammation during the transition to lactation: New adventures with an old flame. *J. Dairy Sci.* 98:6631-6650.
- Campling, R. C. 1970. Physical regulation of voluntary intake. P. 226 in *Physiology of Digestion and Metabolism in the Ruminant*, A. T. Philipson, ed. Newcastle-upon-Tyne, England: Oriel Press. Ltd.
- Choi, B. R., and M. S. Allen. 1999. Intake regulation by volatile fatty acids and physical fill. *S. Afr. J. Anim. Sci.* 29(ISRP):40-41.
- Choi, B. R., D. L. Palmquist, and M. S. Allen. 2000. Cholecystokinin mediates depression of feed intake in dairy cattle fed high fat diets. *Domestic Anim. Endocrinol.* 19:159-175.
- Collier, R. J., R. M. Eley, A. K. Sharma, R. M. Pereira, and D. E. Buffington. 1981. Shade management in subtropical environment for milk yield and composition in Holstein and Jersey cows. *J. Dairy Sci.* 64:844-849.
- Collier, R. J., L. H. Baumgard, R. B. Zimbelman, and Y. Xiao. 2018. Heat stress: Physiology of acclimation and adaptation. *Anim. Front.* 9:12-19.
- Conrad, H. R., A. D. Pratt, and J. W. Hibbs. 1964. Regulation of feed intake in dairy cows. I. Change in importance of physical and physiological factors with increasing digestibility. *Journal of Dairy Science* 47:54.
- Conrad, H. R., C. A. Baile, and J. Mayer. 1977. Changing meal patterns and suppression of feed intake with increasing amounts of dietary non-protein nitrogen in ruminants. *J. Dairy Sci.* 60:1725-1733.
- Dado, R. G., and M. S. Allen. 1994. Variation in feeding behavior characteristics measured continuously among dairy cows in early lactation. *J. Dairy Sci.* 77:132-144.
- Dado, R. G., and M. S. Allen. 1995. Intake limitations, feeding behavior, and rumen function of cows challenged with rumen fill from dietary fiber or inert bulk. *J. Dairy Sci.* 78:118-133.
- Dann, H. M., and B. H. Nelson. 2011. Early lactation diets for dairy cattle—focus on starch. Pp. 46-56 in *Proceedings of the 2011 Cornell Nutrition Conference for Feed Manufacturers*, October 18-20, 2011, East Syracuse, NY. Ithaca, NY: Cornell University.
- Dann, H. M., G. A. Varga, and D. E. Putnam. 1999. Improving energy supply to late gestation and early postpartum dairy cows. *J. Dairy Sci.* 82:1765-1778.
- de Souza, R. A., R. J. Tempelman, M. S. Allen, and M. J. VandeHaar. 2019. Updating predictions of dry matter intake of lactating dairy cows. *J. Dairy Sci.* 102:7948-7960.
- Dehority, B. A. 1969. Pectin-fermenting bacteria isolated from the bovine rumen. *J. Bacteriol.* 99:189-196.

- Demigne, C., C. Yacoub, C. Remesy, and P. Fafoumoux. 1986. Propionate and butyrate metabolism in rat or sheep hepatocytes. *Biochim. Biophys. Acta* 875:535-542.
- DeVries, T. J., and R. M. Gill. 2012. Adding liquid feed to a total mixed ration reduces feed sorting behavior and improves productivity of lactating dairy cows. *J. Dairy Sci.* 95:2648-2655.
- DeVries, T. J., M. A. G. von Keyserlingk, and K. A. Beauchemin. 2003. Short communication: Diurnal feeding pattern of lactating dairy cows. *J. Dairy Sci.* 86:4079-4082.
- DeVries, T. J., M. A. G. von Keyserlingk, and K. A. Beauchemin. 2005. Frequency of feed delivery affects the behavior of lactating dairy cows. *J. Dairy Sci.* 88:3553-3562.
- DeVries, T. J., L. Holtshausen, M. Oba, and K. A. Beauchemin. 2011. Effect of parity and stage of lactation on feed sorting behavior of lactating dairy cows. *J. Dairy Sci.* 94:4039-4045.
- Drackley, J. L., T. H. Klusmeyer, A. M. Trusk, and J. H. Clark. 1992. Infusion of long-chain fatty acids varying in saturation and chain length into abomasum of lactating dairy cows. *J. Dairy Sci.* 75:1517-1526.
- Elliot, J. M., H. W. Symonds, and B. Pike. 1985. Effect on feed intake of infusing sodium propionate or sodium acetate into a mesenteric vein of cattle. *J. Dairy Sci.* 68:1165-1170.
- Felton, C. A., and T. J. DeVries. 2010. Effect of water addition to a total mixed ration on feed temperature, feed intake, sorting behavior, and milk production of dairy cows. *J. Dairy Sci.* 93:2651-2660.
- Forbes, J. M. 2007. *Voluntary Food Intake and Diet Selection in Farm Animals*. Oxfordshire, UK: CAB International.
- Friedman, M. I. 1995. Control of energy intake by energy metabolism. *Am. J. Clin. Nutr.* 62:1096S-1100S.
- Friedman, M. I. 1997. An energy sensor for control of energy intake. *Proc. Nutr. Soc.* 56:41-50.
- Friedman, M. I., and E. M. Stricker. 1976. The physiological psychology of hunger; a physiological perspective. *Psychological Review* 83: 409-431.
- Friend, T. H., and C. E. Polan. 1974. Social rank, feeding behavior, and free stall utilization by dairy cattle. *J. Dairy Sci.* 57:1214-1220.
- Friend, T. H., C. E. Polan, and M. L. McGilliard. 1977. Free stall and feed bunk requirements relative to behavior, production and individual feed intake in dairy cows. *J. Dairy Sci.* 60:108-116.
- Grant, R. J. 1997. Interactions among forages and non-fiber forage sources. *J. Dairy Sci.* 80:1438-1446.
- Grant, R. J., and L. F. Ferraretto. 2018. Silage characteristics and dairy cow feeding behavior. *J. Dairy Sci.* 101:4111-4121.
- Grovum, W. L. 1979. Factors affecting the voluntary intake of food by sheep. 2. The role of distention and tactile input from compartments of the stomach. *Br. J. Nutr.* 42:425.
- Grovum, W. L., and G. D. Phillips. 1978. Factors affecting the voluntary intake of food by sheep. 1. The role of distention, flow-rate of digesta, and propulsive motility of the intestines. *Br. J. Nutr.* 40:323.
- Gualdrón-Duarte, L. B., and M. S. Allen. 2017. Increased anaplerosis of the tricarboxylic acid cycle decreased meal size and energy intake of cows in the postpartum period. *J. Dairy Sci.* 100:4425-4434.
- Gualdrón-Duarte, L. B., and M. S. Allen. 2018. Fuels derived from starch digestion have different effects on energy intake and metabolic responses of cows in the postpartum period. *J. Dairy Sci.* 101:5082-5091.
- Harding, R., and B. F. Leek. 1972. The effect of peripheral influences on gastric centre neuronal activity in sheep. *J. Physiol.* 225:309.
- Hart, K. D., B. McBride, T. F. Duffield, and T. J. DeVries. 2014. Effect of frequency of feed delivery on the behavior and productivity of lactating dairy cows. *J. Dairy Sci.* 97:1713-1724.
- Havekes, C. D., T. F. Duffield, A. J. Carpenter, and T. J. DeVries. 2020a. Effects of molasses-based liquid feed supplementation to a high-straw dry cow diet on feed intake, health, and performance of dairy cows across the transition period. *J. Dairy Sci.* 103:5070-5089.
- Havekes, C. D., T. F. Duffield, A. J. Carpenter, and T. J. DeVries. 2020b. Effects of wheat straw chop length in high-straw dry cow diets on intake, health, and performance of dairy cows across the transition period. *J. Dairy Sci.* 103:254-271.
- Havekes, C. D., T. F. Duffield, A. J. Carpenter, and T. J. DeVries. 2020c. Moisture content of high-straw dry cow diets affects intake, health, and performance of transition dairy cows. *J. Dairy Sci.* 103:1500-1515.
- Hoffman, P. C., K. A. Weigel, and R. M. Wenberg. 2008. Evaluation of equations to predict dry matter intake of dairy heifers. *J. Dairy Sci.* 91:3699-3709.
- Hotter, J. B., J. W. West, M. L. McGillard, and A. N. Pell. 1996. Predicting ad libitum dry matter intake and yields of Jersey cows. *J. Dairy Sci.* 79:912-921.
- Hosseinkhani, A., T. J. DeVries, K. L. Proudfoot, R. Valizadeh, D. M. Veira, and M. A. G. von Keyserlingk. 2008. The effects of feed bunk competition on the feed sorting behavior of close-up dry cows. *J. Dairy Sci.* 91:1115-1121.
- Houseknecht, K. L., C. A. Baile, R. L. Matteri, and M. E. Spurlock. 1998. The biology of leptin: A review. *J. Dairy Sci.* 76:1405-1420.
- Igono, M. O., G. Bjotvedt, and H. T. Sanford-Crane. 1992. Environmental profile and critical temperature effects on milk production of Holstein cows in desert climate. *Int. J. Biometeorol.* 36:77-87.
- Johnson, C., and T. J. DeVries. 2018. Short communication: Associations of feeding behavior and milk production in dairy cows. *J. Dairy Sci.* 101:3367-3373.
- Kammes, K. L., and M. S. Allen. 2012a. Nutrient demand interacts with forage family to affect digestion responses in dairy cows. *J. Dairy Sci.* 95:3269-3287.
- Kammes, K. L., and M. S. Allen. 2012b. Nutrient demand interacts with legume maturity to affect rumen pool sizes in dairy cows. *J. Dairy Sci.* 95:2632-2647.
- Kammes, K. L., and M. S. Allen. 2012c. Rates of particle size reduction and passage are faster for legume compared to cool-season grass, resulting in lower rumen fill and less effective fiber. *J. Dairy Sci.* 95:3288-3297.
- Kennedy, G. C. 1953. The role of depot fat in the hypothalamic control of food intake in the rat. *Proc. R. Soc. B Biol. Sci.* 140:578-592.
- King, M. T. M., R. E. Crossley, and T. J. DeVries. 2016. Impact of timing of feed delivery on the behavior and productivity of dairy cows. *J. Dairy Sci.* 99:1471-1482.
- Langhans, W. 2008. Fatty acid oxidation in the energostatic control of eating—a new idea. *Appetite* 51:446-451.
- Langhans, W., and E. Scharrer. 1992. Metabolic control of eating. *World Rev. Nutr. Diet.* 70:1-67.
- Leek, B. F. 1986. Sensory receptors in the ruminant alimentary tract. Pp. 3-18 in *Control of Digestion and Metabolism in Ruminants*, L. P. Milligan, W. L. Grovum, and A. Dobson, eds. Englewood Cliffs, NJ: Prentice Hall.
- Lehman, F. 1941. Die lehre vom ballast. *Z. Tierphysiol. Tierernährung Futtermittelkunde* 5:155.
- Leonardi, C., and L. E. Armentano. 2003. Effect of quantity, quality, and length of alfalfa hay on selective consumption by dairy cows. *J. Dairy Sci.* 86:557-564.
- Leonardi, C., F. Giannico, and L. E. Armentano. 2005. Effect of water addition on selective consumption (sorting) of dry diets by dairy cattle. *J. Dairy Sci.* 88:1043-1049.
- Longenbach, J. I., A. J. Henrichs, and R. E. Graves. 1999. Feed bunk length requirements for Holstein dairy heifers. *J. Dairy Sci.* 82:99-109.
- Macleod, G. K., P. E. Colucci, A. D. Moore, D. G. Grieve, and N. Lewis. 1994. The effects of feeding frequency of concentrates and feeding sequence of hay on eating behavior, ruminal environment and milk production in dairy cows. *Can. J. Anim. Sci.* 74:103-113.
- Malbert, C. H., and Y. Ruckebusch. 1989. Hyperphagia induced pylorotomy in sheep. *Physiol. Behav.* 45:495.
- Maltecca, C., H. Khatib, V. R. Schutzkus, P. C. Hoffman, and K. A. Weigel. 2006. Changes in conception rate, calving performance, and calf health and survival from the use of crossbred Jersey x Holstein sires as mates for Holstein dams. *J. Dairy Sci.* 89:2747-2754.

- Martineau, R., D. R. Ouellet, E. Kebreab, and H. Lapierre. 2016. Casein infusion rate influences feed intake differently depending on metabolizable protein balance in dairy cows: A multilevel meta-analysis. *J. Dairy Sci.* 99:2748-2761.
- Martinsson, K. 1992. Effects of conservation method and access time on silage intake and milk production in dairy cows. *Grass Forage Sci.* 47:161-168.
- Martinsson, K., and E. Burstedt. 1990. Effects of length of access time to feed and allotment of hay on grass silage intake and production in lactating dairy cows. *Swed. J. Agric. Res.* 20:169-176.
- Mayer, J. 1953. Glucostatic mechanism of regulation of food intake. *N. Engl. J. Med.* 249:13-16.
- Mbanya, J. N., M. H. Aniland, and J. M. Forbes. 1993. The voluntary intake of hay and silage by lactating cows in response to ruminal infusion of acetate or propionate, or both, with or without distension of the rumen by a balloon. *Br. J. Nutr.* 69:713-720.
- McCarthy, M. M., T. Yasui, C. M. Ryan, G. D. Mechor, and T. R. Overton. 2015. Performance of early-lactation dairy cows as affected by dietary starch and monensin supplementation. *J. Dairy Sci.* 98:3335-3350.
- McGrane, M. M. 2000. Carbohydrate metabolism—synthesis and oxidation. Pp. 158-210 in *Biochemical and Physiological Aspects of Human Nutrition*, M. H. Stipanuk, ed. Philadelphia, PA: W.B. Saunders.
- Mertens, D. R. 1987. Predicting intake and digestibility using mathematical models of ruminal function. *J. Anim. Sci.* 64:1548.
- Mertens, D. R. 1994. Regulation of forage intake. Pp. 450-493 in *Forage Quality, Evaluation, and Utilization*, G. C. Fahey, Jr., M. Collins, D. R. Mertens, and L. E. Moser, eds. Madison, WI: American Society of Agronomy, Crop Science Society of America, Soil Science Society of America.
- Miller Cushon, E. K., and T. J. DeVries. 2017. Short communication: Associations between feed push-up frequency, feeding and lying behavior, and milk yield and composition of dairy cows. *Dairy Sci.* 100:2213-2218.
- Minson, D. J. 1963. The effect of pelleting and watering on the feeding value of roughage—a review. *J. Br. Grassland Soc.* 18:39.
- NASEM (National Academies of Sciences, Engineering, and Medicine). 2016. *Nutrient Requirements of Beef Cattle*, 8th rev. ed. Washington, DC: The National Academies Press.
- Niu, M., and K. J. Harvatine. 2018. Short communication: The effects of morning compared with evening feed delivery in lactating dairy cows during the summer. *J. Dairy Sci.* 101:396-400.
- Niu, M., Y. Ying, P. A. Bartell, and K. J. Harvatine. 2014. The effects of feeding time on milk production, total-tract digestibility, and daily rhythms of feeding behavior and plasma metabolites and hormones in dairy cows. *J. Dairy Sci.* 97:7764-7776.
- NRC (National Research Council). 1981. *Effect of Environment on Nutrient Requirements of Domestic Animals*. Washington DC: National Academy Press.
- NRC. 1987. *Predicting Feed Intake of Food-Producing Animals*. Washington, DC: National Academy Press.
- NRC. 1989. *Nutrient Requirements of Dairy Cattle*, 6th rev. ed. Washington, DC: National Academy Press.
- NRC. 1996. *Nutrient Requirements of Beef Cattle*, 7th rev. ed. Washington, DC: National Academy Press.
- NRC. 2001. *Nutrient Requirements of Dairy Cattle*, 7th rev. ed. Washington, DC: National Academy Press.
- Oba, M. 2011. Review: Effects of feeding sugars on productivity of lactating dairy cows. *Can. J. Anim. Sci.* 91:37-46.
- Oba, M., and M. S. Allen. 1999a. Effects of brown midrib 3 mutation in corn silage on dry matter intake and productivity of high yielding dairy cows. *J. Dairy Sci.* 82:135-142.
- Oba, M., and M. S. Allen. 1999b. Evaluation of the importance of the digestibility of neutral detergent fiber from forage: Effects on dry matter intake and milk yield of dairy cows. *J. Dairy Sci.* 82:589-596.
- Oba, M., and M. S. Allen. 2000. Effects of brown midrib 3 mutation in corn silage on productivity of dairy cows fed two concentrations of dietary neutral detergent fiber: 1. Feeding behavior and nutrient utilization. *J. Dairy Sci.* 83:1333-1341.
- Oba, M., and M. S. Allen. 2003a. Dose-response effects of intrauminal infusion of propionate on feeding behavior of lactating cows in early or midlactation. *J. Dairy Sci.* 86:2922-2931.
- Oba, M., and M. S. Allen. 2003b. Effects of corn grain conservation method on feeding behavior and productivity of lactating dairy cows at two dietary starch concentrations. *J. Dairy Sci.* 86:174-183.
- Oba, M., and M. S. Allen. 2003c. Extent of hypophagia caused by propionate infusion is related to plasma glucose concentration in lactating dairy cows. *J. Nutr.* 133:1105-1112.
- Oba, M., and M. S. Allen. 2003d. Intra-ruminal infusion of propionate alters feeding behavior and decreases energy intake of lactating dairy cows. *J. Nutr.* 133:1094-1099.
- Oba, M., and M. S. Allen. 2003e. Hypophagic effects of ammonium are greater when infused with propionate compared to acetate in lactating dairy cows. *J. Nutr.* 133:1100-1104.
- Oldham, J. D. 1984. Protein-energy interrelationships in dairy cows. *J. Dairy Sci.* 67:1090-1114.
- Olofsson, J. 1999. Competition for total mixed diets fed for ad libitum intake using one or four cows per feeding station. *J. Dairy Sci.* 82:69-79.
- Papamandjaris, A. A., D. E. MacDougall, and P. J. Jones. 1998. Medium chain fatty acid metabolism and energy expenditure: Obesity treatment implications. *Life Sci.* 62:1203-1215.
- Patton, R. A. 2010. Effect of rumen-protected methionine on feed intake, milk production, true milk protein concentration, and true milk protein yield, and the factors that influence these effects: A meta-analysis. *J. Dairy Sci.* 93:2105-2118.
- Phelps, A., and B. Drew. 1992. Vastly superior first lactations when heifers fed separately. *Feedstuff's* 64(19): 11.
- Piantoni, P., C. M. Ylloja, and M. S. Allen. 2015. Feed intake is related to changes in plasma non-esterified fatty acid concentration and hepatic acetyl CoA content following feeding in lactating dairy cows. *J. Dairy Sci.* 98:6839-6847.
- Poppi, D. P., D. J. Minson, and J. H. Temouth. 1981. Studies of cattle and sheep eating leaf and stem fractions of grasses: III. The retention time in the rumen of large feed particles. *Aust. J. Agric. Res.* 32:123.
- Quigley, J., III, R. E. James, and M. L. McGilliard. 1986. Dry matter intake in dairy heifers: 2. Equations to predict intake of heifers under intensive management. *J. Dairy Sci.* 69:2863-2867.
- Rabelo, E., R. L. Rezende, S. J. Bertics, and R. R. Grummer. 2005. Effects of pre- and postfresh transition diets varying in dietary energy density on metabolic status of periparturient dairy cows. *J. Dairy Sci.* 88:4375-4383.
- Rabiee, A. R., K. Breinhild, W. Scott, H. M. Golder, E. Block, and I. J. Lean. 2012. Effect of fat additions to diets of dairy cattle on milk production and components: A meta-analysis and meta-regression. *J. Dairy Sci.* 95:3225-3247.
- Rayburn, E. B., and D. G. Fox. 1993. Variation in neutral detergent fiber intake of Holstein cows. *J. Dairy Sci.* 76:544-554.
- Relling, A. E., and C. K. Reynolds. 2007. Feeding rumen-inert fats differing in their degree of saturation decreases intake and increases plasma concentrations of gut peptides in lactating dairy cows. *J. Dairy Sci.* 90:1506-1515.
- Reynolds, C. K. 1995. Quantitative aspects of liver metabolism in ruminants. Pp. 351-372 in *Ruminant Physiology: Digestion, Metabolism, Growth and Reproduction*, W. V. Englehardt, S. Leonhard-Marek, G. Breves, and D. Geisecke, eds. Stuttgart, Germany: Ferdinand Enke Verlag.
- Reynolds, C. K., P. C. Aikman, B. Lupoli, D. J. Humphries, and D. E. Beever. 2003. Splanchnic metabolism of dairy cows during the transition from late gestation through early lactation. *J. Dairy Sci.* 86:1201-1217.
- Ricks, C.A., and R. M. Cook. 1981. Regulation of volatile fatty acid uptake by mitochondrial acyl CoA synthetases of bovine liver. *J. Dairy Sci.* 64:2324-2335.
- Roche, J. R., D. Blache, J. K. Kay, D. R. Miller, A. J. Sheahan, and D. W. Miller. 2008. Neuroendocrine and physiological regulation of intake

- with particular reference to domesticated ruminant animals. *Nutr. Res. Rev.* 21:207-234.
- Roche, J. R., N. C. Friggens, J. K. Kay, M. W. Fisher, K. J. Stafford, and D. P. Berry. 2009. Body condition score and its association with dairy cow productivity, health, and welfare. *J. Dairy Sci.* 92:5769-5801.
- Rockwell, R. J., and M. S. Allen. 2016. Chromium propionate supplementation during the peripartum period interacts with starch source fed postpartum: Production responses during the immediate postpartum and carryover periods. *J. Dairy Sci.* 99:4453-4463.
- Roffler, R. E., J. E. Wray, and L. D. Satter. 1986. Production responses in early lactation to additions of soybean meal to diets containing predominantly com silage. *J. Dairy Sci.* 69:1055-1062.
- Roseler, D. K., D. G. Fox, L. E. Chase, A. N. Pell, and W. C. Stone. 1997. Development and evaluation of equations for the prediction of feed intake for lactating Holstein dairy cows. *J. Dairy Sci.* 80:878-893.
- Russek, M. 1963. Participation of hepatic glucoreceptors in the control of intake of food. *Nature* 197:79-80.
- Sadri, H., G. R. Ghorbani, H. R. Rahmani, A. H. Samie, M. Khorvash, and R. M. Bruckmaier. 2009. Chromium supplementation and substitution of barley grain with corn: Effects on performance and lactation in periparturient dairy cows. *J. Dairy Sci.* 92:5411-5418.
- Sartin, J. L., B. K. Whitlock, and J. A. Daniel. 2011. Neural regulation of feed intake: Modification by hormones, fasting, and disease. *J. Anim. Sci.* 89:1991-2003.
- Scharrer, E. 1999. Control of food intake by fatty acid oxidation and ketogenesis. *Nutrition* 15:704-714.
- Schettini, M. A., E. C. Prigge, and E. L. Nestor. 1999. Influence of mass and volume of ruminal contents on voluntary intake and digesta passage of a forage diet in steers. *J. Anim. Sci.* 77:1896-904.
- Stallings, C. C., G. Kroll, J. C. Kelley, M. L. McGilliard. 1985. A computer ration evaluation program for heifers, dry cows and lactating cows. *J. Dairy Sci.* 68:1015-1019.
- Stocks, S. E., and M. S. Allen. 2012. Hypophagic effects of propionate increase with elevated hepatic acetyl CoA concentration of cows in the early postpartum period. *J. Dairy Sci.* 95:3259-3268.
- Stocks, S. E., and M. S. Allen. 2013. Hypophagic effects of propionic acid are not attenuated during a 3-day infusion in the early postpartum period in Holstein cows. *J. Dairy Sci.* 96:4615-4623.
- Van Soest, P. J. 1965. Symposium on factors influencing the voluntary intake of herbage by ruminants: Voluntary intake in relation to chemical composition and digestibility. *J. Anim. Sci.* 24:834.
- Voelker, J. A., G. M. Burato, and M. S. Allen. 2002. Effects of pretrial milk yield on responses of feed intake, digestion, and production to dietary forage concentration. *J. Dairy Sci.* 85:2650-2661.
- Voelker-Linton, J. A., and M. S. Allen. 2008. Nutrient demand interacts with forage family to affect intake and digestion responses in dairy cows. *J. Dairy Sci.* 91:2694-2701.
- Weber, C., C. Hametner, A. Tuchscherer, B. Losand, E. Kanitz, W. Ofen, S. P. Singh, R. M. Bruckmaier, F. Becker, W. Kanitz, and H. M. Hammon. 2013. Variation in fat mobilization during early lactation differently affects feed intake, body condition, and lipid and glucose metabolism in high-yielding dairy cows. *J. Dairy Sci.* 96:165-180.
- Weigand, E., J. W. Young, and A. D. McGilliard. 1975. Volatile fatty acid metabolism by rumen mucosa from cattle fed hay or grain. *J. Dairy Sci.* 58:1294-1300.
- Weld, K. A., and L. E. Armentano. 2017. The effects of adding fat to diets of lactating dairy cows on total-tract neutral detergent fiber digestibility: A meta-analysis. *J. Dairy Sci.* 100:1766-1779.
- West, J. W., B. G. Mullinix, and J. K. Bernard. 2003. Effects of hot, humid weather on milk temperature, dry matter intake, and milk yield of lactating dairy cows. *J. Dairy Sci.* 86:232-242.
- Weston, R. H. 1996. Some aspects of constraint to forage consumption by ruminants. *Aust. J. Agric. Res.* 47:175-197.
- Wheelock, J. B., R. P. Rhoads, M. J. Vanbaale, S. R. Sanders, and L. H. Baumgard. 2010. Effects of heat stress on energetic metabolism in lactating Holstein cows. *J. Dairy Sci.* 93:644-655.
- Wierenga, H. K. 1990. Social dominance in dairy cattle and the influences of housing and management. *Appl. Anim. Behav. Sci.* 27:201-229.
- Young, B. A. 1983. Ruminant cold stress: Effect on production. *J. Anim. Sci.* 57:1601-1607.
- Zanton, G. I., G. R. Bowman, M. Vazquez-Andn, and L. M. Rode. 2014. Meta-analysis of lactation performance in dairy cows receiving supplemental dietary methionine sources or postruminal infusion of methionine. *J. Dairy Sci.* 97:7085-7101.